

MINI PROJECT 2: SSH BRUTE FORCE SIMULATION

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1. Incident Summary

An SSH Brute Force attack was simulated against an internal server [cite: 225]. The attacker conducted repeated unauthorized login attempts and eventually succeeded in gaining unauthorized access [cite: 226]. This report provides a detailed analysis of the attack mechanics, severity classification, and the necessary incident response actions [cite: 227].

2. Attack Details

Attribute	Value
Attack Type	SSH Brute Force + Unauthorized Access [cite: 229]
Source IP	45.33.32.156 (External / Public) [cite: 229]
Target IP	192.168.1.20 (Internal Server) [cite: 229]
Target Service	SSH — Port 22 [cite: 229]
Attack Window	Start: ~02:14 AM Successful Login: 02:17 AM [cite: 229]
Attack Intensity	50 Failed Attempts in 3 minutes [cite: 229]

3. Severity Classification

The overall severity of this incident is classified as **CRITICAL** [cite: 231, 232].

Factor	Observation	Risk Level
External Attacker	Yes — identified via public IP [cite: 231]	HIGH
Successful Login	Yes — unauthorized access confirmed [cite: 231]	CRITICAL
Time of Attack	02:17 AM (After business hours) [cite: 231]	HIGH
Target Type	Internal production server [cite: 231]	HIGH

4. Immediate Response Actions

- **BLOCK:** Immediately block IP 45.33.32.156 at the perimeter firewall [cite: 234].
- **ISOLATE:** Disconnect server 192.168.1.20 from the network to prevent lateral movement [cite: 235].
- **IDENTIFY:** Verify which specific user account was compromised during the session [cite: 236].
- **RESET:** Force an immediate password reset for the compromised account [cite: 237].
- **AUDIT:** Perform a forensic review of all commands executed post-login [cite: 238].
- **PRESERVE:** Securely capture and backup system logs as evidence [cite: 239].
- **NOTIFY:** Alert the security team and senior management [cite: 240].

5. Escalation & Reporting

Field	Details
Escalation Required?	YES — Immediate escalation [cite: 242]
Escalate To	L2 SOC Analyst / Incident Response Team [cite: 242]
Priority	P1 — Critical [cite: 242]
Justification	Active account compromise confirmed [cite: 242]

6. Security Recommendations

- Enable **fail2ban** to automatically block IP addresses following repeated failed login attempts [cite: 244].
- Disable SSH password-based authentication; enforce **SSH key-based** logins only [cite: 245].
- Implement **IP whitelisting** to restrict SSH access to trusted sources [cite: 246].
- Enforce **Multi-Factor Authentication (MFA)** for all remote administrative access [cite: 247].
- Establish real-time alerting thresholds for failed login attempts [cite: 248].
- Conduct recurring log audits to identify suspicious access patterns early [cite: 249].

7. Conclusion

This incident represents a confirmed SSH Brute Force attack that led to unauthorized access [cite: 251]. The exploit was made possible by weak password controls on the SSH service [cite: 252]. Immediate containment and hardening are mandatory to secure the environment against further compromise and data exfiltration [cite: 253].